

## Advanced (Habanero)

**Q1.** Solve  $x^2 + 5x + 6 = 0$  using the quadratic formula.

- (A)  $x = -2, x = -3$
- (B)  $x = -1, x = -6$
- (C)  $x = 2, x = 3$
- (D)  $x = -2, x = 3$
- (E)  $x = 1, x = -6$

**Q2.** What is the discriminant of  $x^2 + 4x + 4 = 0$ ?

- (A) 0
- (B) 4
- (C) 8
- (D) 12
- (E) 16

**Q3.** How many solutions does  $x^2 - 4x + 4 = 0$  have?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) 4

**Q4.** Solve  $x^2 - 7x + 12 = 0$  using the quadratic formula.

- (A)  $x = 3, x = 4$
- (B)  $x = 2, x = 6$
- (C)  $x = 1, x = 7$
- (D)  $x = 5, x = 8$
- (E)  $x = 3, x = 5$

**Q5.** What is the discriminant of  $x^2 - 3x - 4 = 0$ ?

- (A) 25
- (B) 16
- (C) 9
- (D) 1
- (E) 0

**Q6.** How many solutions does  $x^2 + 2x + 2 = 0$  have?

- (A) 0
- (B) 1
- (C) 2
- (D) 2 complex
- (E) 2 real

**Q7.** Solve  $x^2 + 2x - 6 = 0$  using the quadratic formula.

- (A)  $x = -1 + \sqrt{7}, x = -1 - \sqrt{7}$
- (B)  $x = -1 + \sqrt{5}, x = -1 - \sqrt{5}$
- (C)  $x = 1 + \sqrt{3}, x = 1 - \sqrt{3}$
- (D)  $x = 2 + \sqrt{2}, x = 2 - \sqrt{2}$
- (E)  $x = 3 + \sqrt{1}, x = 3 - \sqrt{1}$

**Q8.** What is the discriminant of  $x^2 - 2x - 3 = 0$ ?

- (A) 13
- (B) 12
- (C) 11
- (D) 10
- (E) 9

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**Open Ended Questions (FRQ)**

**Q9.** Solve  $x^2 + 5x + 5 = 0$  and determine the number and type of solutions. Show all work.

**Q10.** Find the discriminant of  $x^2 - 4x - 5 = 0$ . Then, determine the number and type of solutions. Show all work.

**Chef's Tips**

- Check calculations carefully
- Emphasize the importance of the discriminant
- Provide examples of real-world applications